# MATHEMATICS SSC-II

## Science Group | 2nd Annual Examination 2026

## SECTION A (Marks 15)

Time allowed: 20 minutes | Total marks: 15

Roll number: _____ Candidate signature: _____

Q.1 Choose the correct option for each question. Each question carries one mark.

1. The multiplicative inverse of 3 + 4i is:

A. (3 + 4i)/25 B. (3 − 4i)/25 C. 3 − 4i D. 1/3 − 1/(4i)

2. If $z + \bar{z} = 14 + 0i$, where $\bar{z}$ is the conjugate of z, then the real part of z is:

A. −7 B. 0 C. 7 D. 14

3. For what value of k will the roots of $2x^2 + kx + 8 = 0$ be equal?

A. −8 B. 8 C. ±8 D. ±64

4. If f(x) = x + 4 and g(x) = 3x − 2, then (f ∘ g)(3) is:

A. −11 B. −9 C. 9 D. 11

5. The vertex of the graph y = |x − 3| + 2 is:

A. (−3, 2) B. (3, −2) C. (3, 2) D. (−3, −2)

6. If A = [5 3; −1 4] and B = [2 −1; 3 −2], then the transpose of A − B is:

A. [3 4; -4 2] B. [3 -4; 4 2] C. [2 4; -4 3] D. [2 -4; 4 3]

7. If a = 3i + 6j and b = i + mj are perpendicular, then m is:

A. −2 B. 2 C. −1/2 D. 1/2

8. Simplified form of $((x + y)^2 + (x - y)^2) / (8(x^2 + y^2))$ is:

A. 1/4 B. 1/2 C. 2 D. 4

9. $(x^2 - 1)/(x - 1)$ =

A. x − 1 B. x + 1 C. $x^2 + 1$ D. 1

10. If the product of two consecutive even numbers is 48, the quadratic equation formed is:

A. $x^2 + 2x + 48 = 0$ B. $x^2 + 2x - 48 = 0$ C. $x^2 - x - 48 = 0$ D. $x^2 + x + 48 = 0$

11. The graphs of y = 2x + 1 and y = 2x − 4 will be:

A. Intersecting B. Parallel C. Perpendicular D. Represent same line

12. If a = 9i − 7j and b = 4i + 5j, then the magnitude of a − b is:

A. 13 B. 14 C. 15 D. 16

13. A vertical pole of 15 m height is supported by a wire. The distance between the pole and the point supporting the wire on the ground is 8 m. The length of the wire is:

A. 13 B. 15 C. 17 D. 23

14. A semicircular bridge has a span of 20 m. The length of the chord joining the two ends of the bridge is:

A. 5 m B. 10 m C. 15 m D. 20 m

15. A card is drawn from 52 cards. The probability of getting a king is:

A. 1/52 B. 1/26 C. 1/13 D. 4/13

# MATHEMATICS SSC-II

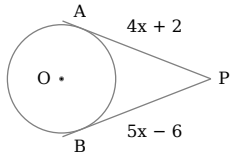
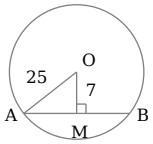
## Science Group | 2nd Annual Examination 2026

## SECTION B (Marks 36)

Time allowed: 2 hours 40 minutes (Sections B and C) Total marks: 60 (B + C)

Q.2 Solve the following parts. (9 × 4 = 36)
Choose either alternative in each part. Each alternative carries 4 marks.

Use of graph paper is not necessary. Candidates can make their own grid on the answer book.

| | |
|---|---|
| (i) [4]<br>If $z_1 = 2 + 3i$ and $z_2 = 1 - 2i$, verify that the conjugate of $z_1z_2$ equals the product of the conjugates of $z_1$ and $z_2$. | OR [4]<br>A weather report states that the probability of rain on Monday is 0.4 and the probability of rain on Tuesday is 0.6. Assuming these events are independent, calculate the probability that it will be rainy on both Monday and Tuesday. |
| (ii) [4]<br>For what value of k do the roots of the following equation differ by 5?<br><br>$x^2 + kx + 6 = 0$ | OR [4]<br>In the given figure, find x. Also find the lengths of tangents PA and PB.<br>  |
| (iii) [4]<br>For the following matrices, find (A + B)(A − B).<br><br>A = [7 10; 4 6], B = [5 1; 2 6] | OR [4]<br>In the given figure, find the length of chord AB.<br>  |
| (iv) [4]<br>Given $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $g(f(x))$. Also find $g(f(3))$. | OR [4]<br>Two sides of a triangular field are 24 m and 18 m, and the angle between them is 50°. Find the area of the field. |
| (v) [4]<br>Plot the graph of the following absolute-value equation by making a table of at least five values.<br><br>$y = \|2x + 4\|$ | OR [4]<br>The position vectors of points A and B are (2, −1) and (8, 7), respectively. Find vector AB. Also find its magnitude. |

# MATHEMATICS SSC-II

**Science Group | 2nd Annual Examination 2026**

**SECTION B (Marks 36)**

Time allowed: 2 hours 40 minutes (Sections B and C) Total marks: 60 (B + C)

Q.2 Solve the following parts. (9 × 4 = 36)
Choose either alternative in each part. Each alternative carries 4 marks.

Use of graph paper is not necessary. Candidates can make their own grid on the answer book.

| | |
|---|---|
| (vi) [4]<br>Simplify the following expression:<br><br>$[(x^2 - 4)/(x^2 - 2x + 1)] \div [(x + 2)/(x - 1)]$ | OR [4]<br>Draw the graph of the following equation by making a table of at least five values.<br><br>$y = 3x^2 - 3$ |
| (vii) [4]<br>A school canteen sells sandwiches and juice. Let x represent sandwiches and y represent juices. The inequalities formed are $x + y \leq 10$ and $2x + y \leq 14$, with $x, y \geq 0$. Graph the inequalities and shade the solution. | OR [4]<br>Solve the following equation by completing the square:<br><br>$2x^2 + 8x - 10 = 0$ |
| (viii) [4]<br>Find the multiplicative inverse of the following matrix:<br><br>$A = [5\ -3;\ -4\ 7]$ | OR [4]<br>Construct a circle of radius 3 cm. Draw two tangents from a point P which is 6 cm away from the centre of the circle. |
| (ix) [4]<br>Abdullah rolled a fair die once. What is the probability of getting a number greater than 2, and the probability of getting a number less than 3? Also add both probabilities. | OR [4]<br>Solve the following equation:<br><br>$(x + 2)/(x - 1) = 2/(x + 2)$, where $x \neq -2, 1$. |

# MATHEMATICS SSC-II

**Science Group | 2nd Annual Examination 2026**

**SECTION C (Marks 24)**

Time allowed: 2 hours 40 minutes (Sections B and C) Total marks: 60 (B + C)

Solve all three questions, choosing either alternative in each. (3 × 8 = 24)
Use of graph paper is not necessary. Candidates can make their own grid on the answer book.

Q.3 [8]
Solve the following simultaneous linear equations with complex coefficients:

$(2 + i)x + (1 - i)y = 5 + i$

$(1 + i)x + (2 - i)y = 4 + 3i$

OR [8]
A surveyor stands at point A on one bank of a river. A tree T is directly opposite A. He walks 40 m along the bank to point B. The angle ABT is 52°.
(a) Draw a suitable diagram.
(b) Calculate the width of the river AT.
(c) Find the distance BT.

Q.4 [8]
Solve the following equations graphically on the same coordinate plane:

$y = x^2 - 4$; $y = x + 2$

OR [8]
A rectangular field is 20 m long and 14 m wide. A uniform path is constructed outside the field. The total area of the field and path is 720 $m^2$.
(a) Form a quadratic equation for the problem.
(b) Find the width of the path.

Q.5 [8]
The following table shows the data of monthly articles made in a factory. Use this data and draw an ogive.

| Class interval | Frequency |
|---|---|
| 100 - 200 | 5 |
| 200 - 300 | 8 |
| 300 - 400 | 6 |
| 400 - 500 | 9 |
| 500 - 600 | 12 |

OR [8]
Construct two tangents to a circle of radius 4 cm, such that the angle between the two tangents is 60°.

## SOURCE FACSIMILE - SECTION B

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# MATHEMATICS SSC–II

## Science Group

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**Time allowed: 2:40 Hours** **Total Marks Sections B and C: 60**

### SECTION – B (Marks 36)

**Q. 2 Solve the following parts** **( 9 x 4 = 36 )**

سوال نمبر 2 تمام اجزا حل کریں۔

**(Use of graph paper is not necessary. Candidates can make their own grid on answer book)**

**(i)** If $Z_1 = 2 + 3i$ and $Z_2 = 1 - 2i$
Then verify that
$Z_1 \cdot Z_2 = \overline{Z_1} \cdot \overline{Z_2}$
04

OR

$Z_1 = 1 - 2i$ اور $Z_2 = 2 + 3i$ ہوں تو ثابت کریں کہ
$\overline{Z_1 \cdot Z_2} = \overline{Z_1} \cdot \overline{Z_2}$

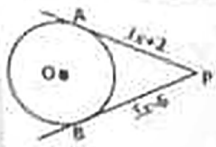
A weather repon states that the probability of rain on Monday is 0.4 and the probability of rain on Tuesday is 0.6. Assume these events are independent. Calculate the probability that it will be rain on both Monday and Tuesday
04

موسم کی رپورٹ کے مطابق پیر کے دن بارش ہونے کا امکان 0.4 اور منگل کے دن بارش ہونے کا امکان 0.6 ہے۔ فرض کریں کہ یہ واقعات آزاد ہیں۔ دونوں دن بارش ہونے کا امکان معلوم کریں۔

**(ii)** For What value of $k$ the roots of the following equation differ by 5.
$x^2 + kx + 6 = 0$
04

OR

In the given figure find the value of $x$. Also find lengths of tangents $\overline{PA}$ and $\overline{PB}$
04




دیئے گئے شکل میں $x$ کی قیمت معلوم کریں۔ نیز $\overline{PA}$ اور $\overline{PB}$ مماسوں کی لمبائیں بھی معلوم کریں۔

**(iii)** For the following matrices find $(A + B)(A - B)$
$A = \begin{bmatrix} 7 & 10 \\ 4 & 6 \end{bmatrix}$ $B = \begin{bmatrix} 5 & 1 \\ 2 & 6 \end{bmatrix}$
04

OR

In the given figure find the length of chord $\overline{AB}$
04




دیئے گئے میٹرکس کے لیے $(A + B)(A - B)$ معلوم کریں۔

دیئے گئے شکل میں $\overline{AB}$ کی لمبائی معلوم کریں۔

**(iv)** Given $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $g(f(x))$ Also find $g(f(3))$
04

OR

Two sides of a triangular field are 24 m and 10 m and the angle between them is 50°. Find the area of the field
04

$g(x) = x^2 - 3$ اور $f(x) = 2x + 1$ دیئے گئے ہیں۔ $g(f(x))$ معلوم کریں۔ نیز $g(f(3))$ بھی معلوم کریں۔

ایک مثلثی کھیت کے دو اضلاع کی لمبائیاں 24 m اور 10 m ہیں اور ان کے درمیان زاویہ 50° ہے۔ کھیت کا رقبہ معلوم کریں۔

**(v)** Plot the graph of the following absolute equation by making a table of at least five values
$y = |2x + 4|$
04

OR

The position vectors of points $A$ and $B$ are $[2, -1]$ and $[8, 7]$ respectively. Find vector $\overrightarrow{AB}$. Also find its magnitude
04

دیئے گئے مطلق مساوات کا گراف بنائیں (کم از کم پانچ قیمتوں کا جدول بنائیں)

نقطہ A اور B کے مقام سمتیہ بالترتیب $[2, -1]$ اور $[8, 7]$ ہیں۔ سمتیہ $\overrightarrow{AB}$ معلوم کریں۔ نیز اس کی مقدار (magnitude) بھی معلوم کریں۔

**(vi)** Simplify the following expression
$\frac{x^2 + 4}{x^2 - 2x + 4} \div \frac{x + 2}{x - 3}$
04

OR

Draw the graph of the following equation, by making table of at least 5 values
$y = 3x^2 - 3$
04

درج ذیل کو سادہ کریں۔

کم از کم پانچ قیمتوں کا جدول بنا کر درج ذیل کا گراف بنائیں۔

**(vii)** A school canteen sells sandwiches and juice. Let $x$ represent sandwiches and $y$ represent juices. The inequalities formed are $x + y \leq 10$ and $2x + y \leq 14$, with $x, y \geq 0$. Graph the inequalities and shade the solution
04

OR

Solve the following equation by completing square method
$2x^2 + 8x - 10 = 0$
04

ایک سکول کی کینٹین سینڈوچز اور جوس فروخت کرتی ہے۔ فرض کریں $x$ سینڈوچز اور $y$ جوس کی تعداد ہے۔ بنائی گئی عدم مساوات $x + y \leq 10$ اور $2x + y \leq 14$ ہیں جہاں $x, y \geq 0$ ہے۔ عدم مساوات کا گراف بنائیں اور حل کو ظاہر کریں۔

**(viii)** Find the multiplication inverse of the following Matrix
$A = \begin{bmatrix} 5 & -3 \\ -4 & 7 \end{bmatrix}$
04

OR

Construct a circle of radius 3 cm. Draw two tangents from a point $P$, which is 6cm away from the centre of circle
04

درج ذیل میٹرکس کا ضربی معکوس معلوم کریں۔

3 cm رداس کا دائرہ بنائیں۔ مرکز سے 6cm کے فاصلے پر نقطہ P سے دائرے کے دو مماس کھینچیں۔

**(ix)** Abdullah rolled a fair dice once. What is the probability of getting a number greater than 2 and find probability of getting a number less than 3. Also add both the probabilities
04

OR

Solve the following equation
$\frac{x+2}{x-1} = \frac{3}{x+2}$ where $x \neq -2, 1$
04

عبداللہ نے ایک منصفانہ ڈائس (fair dice) کو ایک مرتبہ پھینکا۔ 2 سے زیادہ آنے کا امکان معلوم کریں۔ نیز 3 سے کم آنے کا امکان بھی معلوم کریں۔ پھر دونوں امکانات کو جمع کریں۔

درج ذیل مساوات کو حل کریں۔

## SOURCE FACSIMILE - SECTION C

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## SECTION – C (Marks 24)

**Note:** Solve the following Questions. (3 x 8 = 24)

تمام سوالات حل کریں۔

**(Use of graph paper is not necessary. Candidates can make their own grid on answer book)**

**Q.3** Solve the following simultaneous linear equations with complex coefficients.

$$(2+i)x + (1-i)y = 5+i$$

$$(1+i)x + (2-i)y = 4+3i$$

ذیل معادلاتی سر راستے سے (کا) کے ساتھ مساوات کو حل کریں۔

08

OR

A surveyor stands at point $A$ on one bank of a river. A tree $T$ is directly opposite $A$. He walks 40 m along the bank to point $B$. The angle $ABT$ is $52^\circ$.

(a) Draw a suitable diagram.

(b) Calculate the width of the river $AT$.

(c) Find the distance $BT$.

ایک سروے دریا کے ایک کنارے پر نقطہ $A$ پر کھڑا ہے۔ ایک درخت $T$، نقطہ $A$ کے عین مخالف ہے۔ وہ کنارے کے ساتھ 40 m چل کر نقطہ $B$ تک جاتا ہے۔ زاویہ $ABT$ ہے۔

(الف) مناسب شکل بنائیں۔

(ب) دریا کی چوڑائی $AT$ معلوم کریں۔

(ج) $BT$ کا فاصلہ بھی معلوم کریں۔

08

**Q.4** Solve the following equations graphically, on the same coordinate plane.

$$y = x^2 - 4 \quad , \quad y = x + 2$$

دیئے گئے مساوات کو گرافی طریقے سے ایک ہی محدد کی مستوی پر حل کریں۔

08

OR

A rectangular field is 20 m long and 14 m wide. A uniform path is constructed outside the field. The total area of the field and path is $720\ m^2$.

(a) Form a quadratic equation for the Problem.

(b) Find the width of the path.

ایک مستطیل کھیت کی لمبائی 20 m اور چوڑائی 14 m ہے۔ کھیت کے باہر یکساں چوڑائی کا راستہ بنایا گیا ہے۔ کھیت اور راستے کا کل رقبہ $720\ m^2$ ہے۔

(الف) راستے کے لیے دی گئی مساوات بنائیں۔

(ب) راستے کی چوڑائی معلوم کریں۔

08

**Q.5** The following table shows the data of monthly articles made in a factory. Use this data and draw an ogive.

| Class interval | Frequency $f$ |
|---|---|
| 100 – 200 | 5 |
| 200 – 300 | 8 |
| 300 – 400 | 6 |
| 400 – 500 | 9 |
| 500 – 600 | 12 |

دیئے گئے جدول میں کسی کارخانے میں ماہانہ تیار ہونے والی اشیا کی معلومات ظاہر کرتا ہے۔ ان معلومات کی مدد سے تعدادی گرافک ادائیہ (ogive) بنائیں۔

08

OR

Construct two tangents to a circle of radius 4 cm, such that the angle between the two tangents is $60^\circ$.

ایک دائرہ بنائیں جس کی رداس کی لمبائی 4 cm ہے۔ اس دائرے پر دو ایسے مماس بنائیں کہ ان کے درمیان زاویہ $60^\circ$ ہو۔

08

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